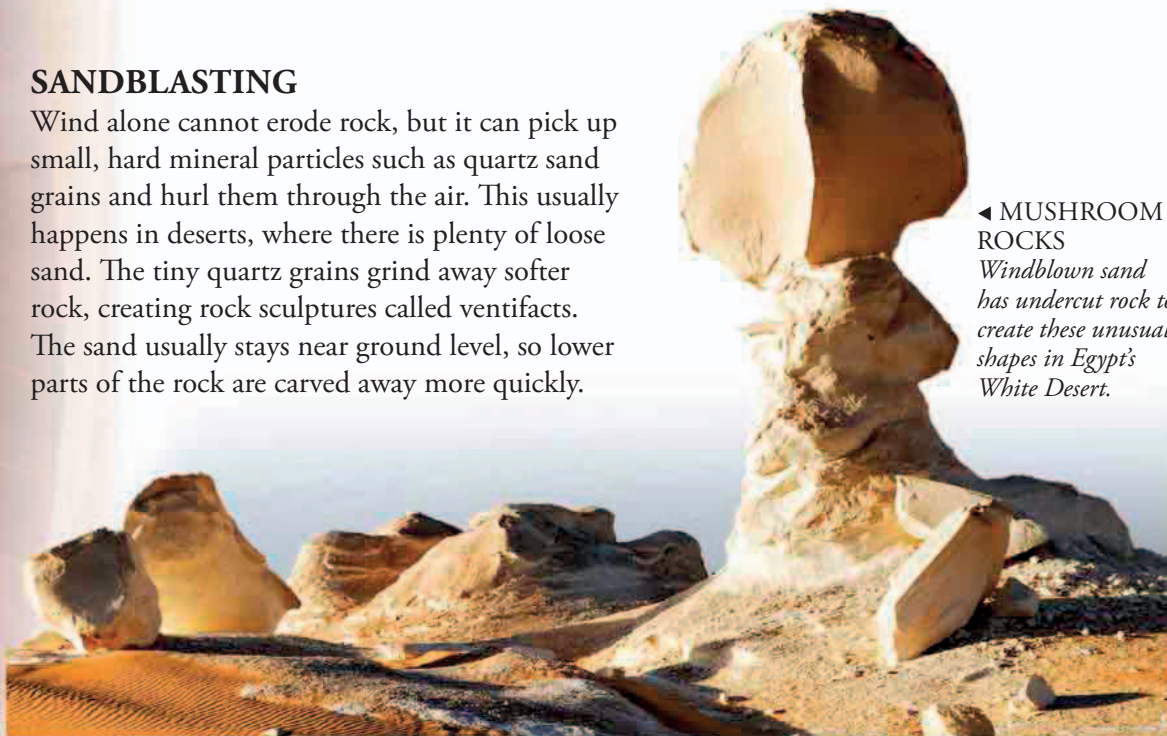




SANDBLASTING

Wind alone cannot erode rock, but it can pick up small, hard mineral particles such as quartz sand grains and hurl them through the air. This usually happens in deserts, where there is plenty of loose sand. The tiny quartz grains grind away softer rock, creating rock sculptures called ventifacts. The sand usually stays near ground level, so lower parts of the rock are carved away more quickly.



◀ MUSHROOM ROCKS

Windblown sand has undercut rock to create these unusual shapes in Egypt's White Desert.

ASH CLOUD

Some volcanic eruptions blast rocky debris high into the air. The wind may carry the finest ash particles around the globe, but the bigger rocks fall to the ground, where they settle as a deposit called tephra. The heavier rock falls first, while lighter particles are carried farther away on the wind. Volcanic ash that simply pours down the flank of a volcano settles as a mixture of big and small particles called ignimbrite.



▲ ASH LAYERS

The rocks of the Painted Hills in Oregon are solidified layers of volcanic ash.



WINDSWEPT DUNES

Wind sweeps sand into hills called dunes, especially in deserts and on seashores. The dunes often creep slowly in the direction that the wind is blowing, as sand grains blow up and over the dune crests. This creates sloping layers of sand, which are called dune cross-bedding. The sand grains themselves are smashed together over and over again, giving them a rounded, "frosted" look.

► MOVED BY ICE

Tiny mineral particles are mixed with big, sharp-edged rock fragments in this lump of glacial till.



GLACIER ICE

In high mountains and polar regions, slowly moving glacier ice freezes onto the rock that it is flowing over and rips away rock fragments. These become embedded in the moving ice, making it grind away more rock. The glacier carries the mixture of sharp fragments and finely ground "rock flour" within the ice flow, so it is not sorted or rounded off by further erosion. Whenever the ice melts, the rock debris is deposited as a sediment known as till, or boulder clay.